

Technical Document #695: Natural Language Processing - Comprehensive Overview

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Named entity recognition (NER) identifies and classifies named entities in text into predefined categories such as person names, organizations, and locations.

Word embeddings represent words as dense vectors in continuous space. Word2Vec, GloVe, and FastText are popular word embedding techniques.

Tokenization is the process of breaking text into individual units called tokens. Subword tokenization methods like BPE handle out-of-vocabulary words gracefully.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Key Takeaways:

- Sentiment analysis determines the emotional tone behind a body of text.
- Language models predict the probability of a sequence of words.
- Machine translation converts text from one language to another.